

# Elicit vs. Teach, Mechanistically: Results on the TinyStories-1B Twins

MARS V — Mechanistic Understanding of Elicitation vs. Teaching

Working results document, September 2026

## Setup

Two identical TinyStories-1B twins (same architecture, same pretraining on children’s stories, zero arithmetic) are fine-tuned on the same target task — signed 1–4-digit addition/subtraction in natural language, **What is the difference between 8 and 5784?  $\rightarrow$  -5776** — with one byte-held protocol (LoRA  $r=512$ ,  $\alpha=32$ , lr  $3.53\times 10^{-4}$ , batch 128, seed 316, eps/k convergence stop, frozen D\_algo\_bare,  $n = 10^2 \dots 10^6$ ). The arms differ *only* in the parent checkpoint:

- **Teach twin:** the blank base (evt-ts1b-base). The capability is absent.
- **Elicit twin: TS1B-latent** (evt-ts1b-op-bridge-mix) — the blank base plus (i) a symbol-only arithmetic install ( $23 + 45 = 68$ , 1M examples, full FT) and (ii) an answer-free word $\leftrightarrow$ symbol binding dose (rewriting **What is the sum of 621 and 5068?  $\leftrightarrow$  621 + 5068**, rehearsed 1:1 with already-seen symbol rows). Every installation is answer-free and leakage-controlled (Table 1). The capability is latent.

**Measurement vocabulary.** *Circuit* = top-32 of 528 nodes (512 attention heads + 16 MLP outputs) by attribution score (gradient of the clean/corrupt logit-diff  $\times$  activation delta, 128–256 pairs); overlaps are Jaccard@32 against a chance floor of 0.031 and a per-map split-half ceiling. *Weight write*  $= \Delta W = \text{scaling} \cdot BA$  per LoRA module (exact; singular values via thin QR); total write  $\|\Delta W\|_F$ , relative write  $\|\Delta W\|/\|W\|$ , travel  $\|\Delta W_t\|_F$  and speed  $\|\Delta W_t - \Delta W_{t-1}\|_F/\Delta t$  from adapter snapshots (the measurable form of “gradient strength” under AdamW, whose updates are  $\sqrt{v}$ -normalised).

| Construction step / check                       | value        | control                 |
|-------------------------------------------------|--------------|-------------------------|
| symbol install: EM on $23 + 45 =$               | 0.67–0.73    | blank: 0.00             |
| corpus census: “sum” in 439M words              | $21\times$   | “equals” 17, “minus” 57 |
| binding dose: NL $\rightarrow$ op rewrite exact | 0.484        | blank + dose: 0.000 EM  |
| sum-only dose on <i>difference</i> questions    | writes “+”   | per-word binding        |
| <b>direct NL exact match of the parent</b>      | <b>0.000</b> | = blank twin            |

Table 1: The latent parent is behaviorally identical to the blank twin on the target task. Everything below is about what differs inside.

## Results

### R1 The learning-curve signature flips.

**Metric.** Prequential EDL per label token (nats) and held-out exact match, as a function of dataset size  $n$ , identical protocol, parent varied.

**Result.** Figure 1, Table 2. Elicit: EDL strictly monotone,  $4.53 \rightarrow 0.092$ ; EM 0.54 at  $n=3,162$ , 0.98 at 1M. Teach: EDL down-up-down with the hump peaking near  $n=215K$ ; EM 0.093 at 1M. EM-0.5 threshold:  $\sim 3 \times 10^3$  vs  $> 10^6$  ( $> 300\times$ ). A third arm (blank + format-only dose) keeps its hump.

**Elicit vs. teach.** Eliciting spends every example connecting an existing capability (diminishing cost from the first rung); teaching must build the capability, producing increasing returns mid-run. The flip is the paper’s causal-intervention claim reproduced on one substrate, with a parent whose latency is itemised rather than assumed.

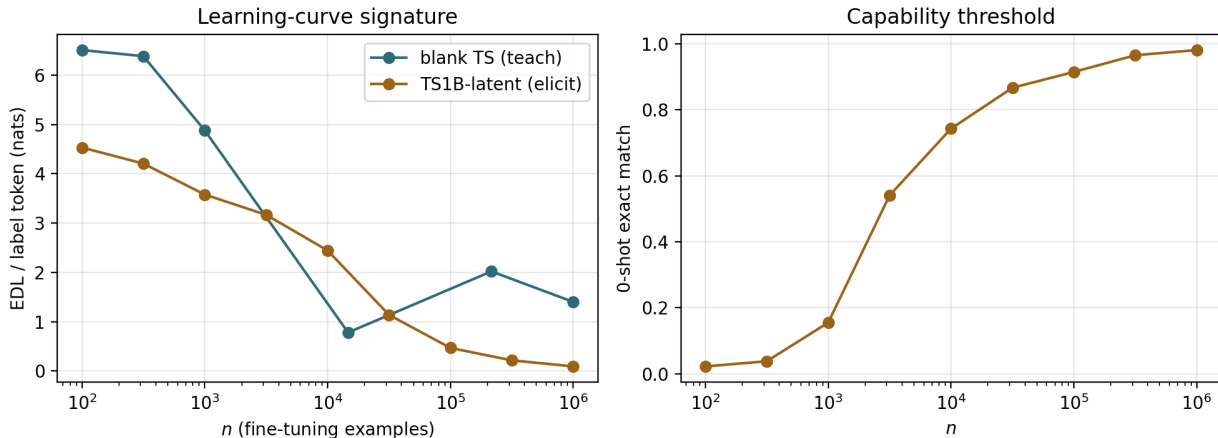


Figure 1: EDL/token (left) and 0-shot EM (right) vs  $n$ ; teal = teach twin, gold = elicit twin.

| $n$       | elicit       |              | teach                 |          |
|-----------|--------------|--------------|-----------------------|----------|
|           | EDL/tok      | EM           | EDL/tok               | EM       |
| 100       | 4.533        | 0.023        | 6.511                 | 0.007    |
| 316       | 4.208        | 0.038        | 6.387                 | 0.033    |
| 1,000     | 3.579        | 0.155        | $\sim 4.9$            | $\sim 0$ |
| 3,162     | 3.168        | <b>0.541</b> | $\downarrow$          | $\sim 0$ |
| 10,000    | 2.444        | 0.743        | min $\sim 0.8$        | $\sim 0$ |
| 31,623    | 1.137        | 0.867        | $\uparrow$            | $\sim 0$ |
| 100,000   | 0.467        | 0.915        | $\uparrow$            | $\sim 0$ |
| 316,228   | 0.215        | 0.966        | peak $\sim 2.0$ @215K | $\sim 0$ |
| 1,000,000 | <b>0.092</b> | <b>0.981</b> | 1.40                  | 0.093    |

Table 2: Per-rung values (EDL nats/label token; EM on 1,024 held-out problems).

**R2 The elicited circuit is the parent’s installed circuit; the taught circuit matches nothing upstream.**

**Metric.** Jaccard@32 between attribution circuits (chance 0.031; ceiling = split-half self-agreement of the map, 0.684 for the elicited child).

**Result.** Table 3. Parent (installed engine)  $\leftrightarrow$  elicited child: 0.455–0.524 across eval surfaces ( $\approx 70\%$  of ceiling), top-4 nodes identical and identically ordered (`mlp:15,14,13,12`). Blank parent: no measurable circuit (logit-diff  $-0.33$  even 16-shot; its map is noise), so the taught child has nothing to match.

**Elicit vs. teach.** Elicitation re-weights machinery that was already there; teaching constructs machinery where there was none. “Reuse” is measured, not inferred.

| Comparison                                                              | J@32                            | chance | ceiling       |
|-------------------------------------------------------------------------|---------------------------------|--------|---------------|
| installed engine $\leftrightarrow$ elicited child (op / bridge surface) | 0.455–0.488 / 0.524             | 0.031  | 0.684         |
| blank parent $\leftrightarrow$ taught child                             | undefined (parent map is noise) | –      | –             |
| elicited child @3K $\leftrightarrow$ @1M                                | 0.391                           | 0.031  | 0.684         |
| elicited child @1M $\leftrightarrow$ taught child @1M                   | 0.231 (Spearman $-0.11$ )       | 0.031  | 0.684 / 0.561 |

Table 3: Circuit-membership overlaps. All maps pass the performing-regime guard.

### R3 Same task, same substrate, different machine.

**Metric.** Layer distribution of the top-16 attribution nodes; J@32 and score Spearman between the two children.

**Result.** Figure 2. Elicited child: mid/late MLP backbone (layers 5–15) plus the layer-7 attention trio — the installed engine’s own profile. Taught child: eight layer-0 attention heads over raw digit tokens feeding a few late MLPs. Overlap 0.231, score correlation  $-0.11$ .

**Elicit vs. teach.** Because both children share the identical pretrained substrate, the mechanism difference is caused by the learning regime alone — the cross-model version of this finding had a pretraining confound; this one does not.

### R4 The elicited circuit exists before training; the taught circuit is built mid-run — through the EDL hump.

**Metric.** Jaccard@32 of each training snapshot’s circuit against the run’s final circuit.

**Result.** Figure 3. Elicit: 0.488 at step 1 ( $16\times$  chance), a brief interface-wiring transient (dip to  $\sim 0.42$  with a strong map at step 154), locked from  $\sim$ step 1.5K of 11.5K (0.561; 82% of ceiling, 97% of the J@64 ceiling by 8.4K). Teach: noise floor ( $\approx 0.22$ – $0.28$ ) until  $\sim$ step 1.3K, then  $0.27 \rightarrow 0.44$  over steps 1.3K–5.4K — the region where its EDL peaks.

**Elicit vs. teach.** Elicitation finds and wires; teaching constructs. The elicited circuit finishes stabilising before the taught one begins to exist, and the teaching hump is literally the circuit being assembled.

### R5 The words reach the engine only when the binding is installed.

**Metric.** Cross-format activation matching: per node, mean cosine between the activation on the NL rendering and on the symbol rendering of the *same* problem, minus the mismatched-pair baseline (“does What is the sum of 621 and 5068? drive the node into the same problem-specific state as  $621 + 5068 = ?$ ”).

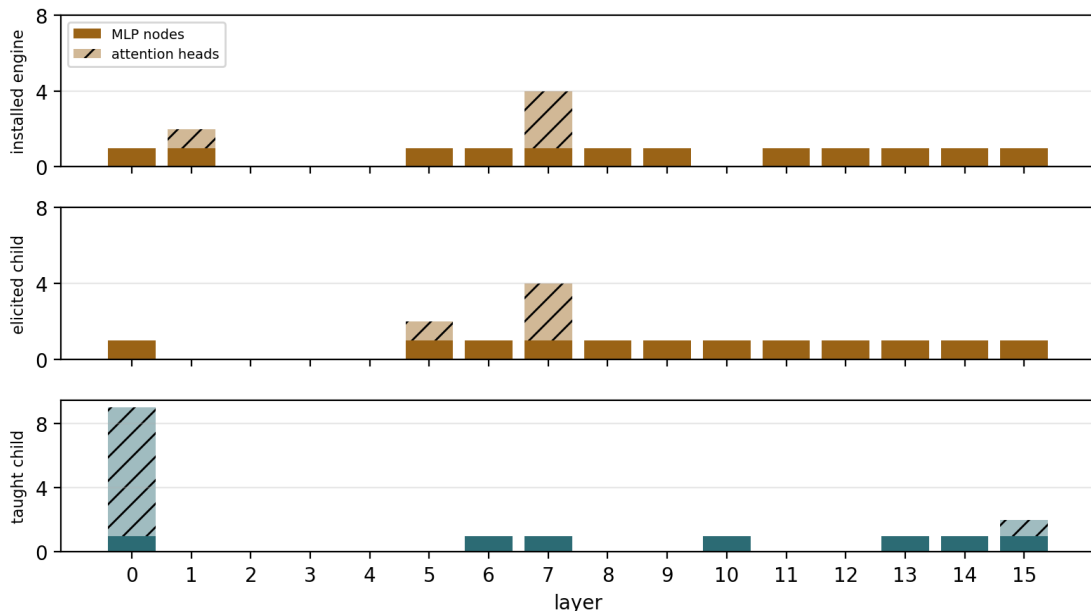


Figure 2: Top-16 node layer histograms (solid = MLP, hatched = attention) for the installed engine, the elicited child, and the taught child.

**Result.** Engine without binding: MLP index  $\approx 0.00$ – $0.01$  at every layer. TS1B-latent (binding installed):  $+0.07$ – $0.13$  across layers 7–15,  $\sim 10\times$ . Attention-head indices are confounded by shared digit tokens and are excluded.

**Elicit vs. teach.** This is what “latent” means internally: before any target training, NL input already activates the arithmetic MLPs of the elicit parent, while the teach parent has no arithmetic MLPs to activate. It also shows the binding dose did what it claimed — installed an access route, not the task.

## R6 The latent capability can be reached with frozen weights; the absent one cannot.

**Metric.** Exact match on the target task under zero-training interventions: (a) self-chain — the model rewrites the question itself, then computes its own rewrite; (b) donor patching — the fine-tuned child’s activations at the 32 circuit nodes written into the parent (mean vector, or per-prompt states); random-node controls.

**Result.** Figure 4, Table 4. Elicit parent: direct 0.000  $\rightarrow$  self-chain **0.328** ( $= 0.484 \times 0.668$ ; the composition law  $\text{parse} \times \text{compute}$  held on four model variants); per-prompt states 0.125 (random nodes: 0.000); mean vector: format only. Teach side: blank + same doses 0.000; blank + chain 0.000 (a parser without an engine); taught donor into blank twin 0.000 in every condition.

**Elicit vs. teach.** Elicit = one composition step or one state-patch away from the target; teach = nothing underneath to switch on. The teach-side nulls double as the leakage control for the whole construction.

## R7 Teaching writes far more into the weights, and buys far less per unit written.

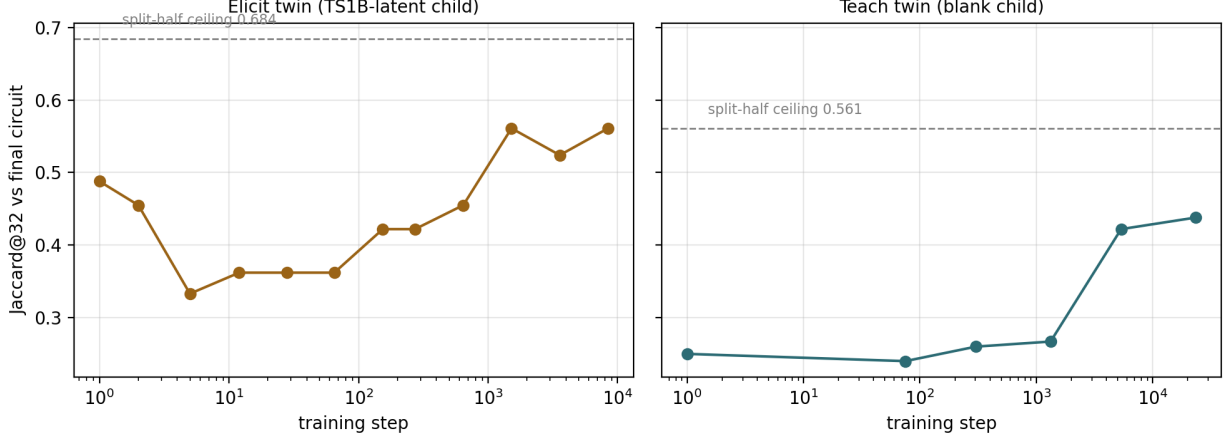


Figure 3: Circuit formation, elicit (left) vs teach (right); dashed = split-half ceiling.

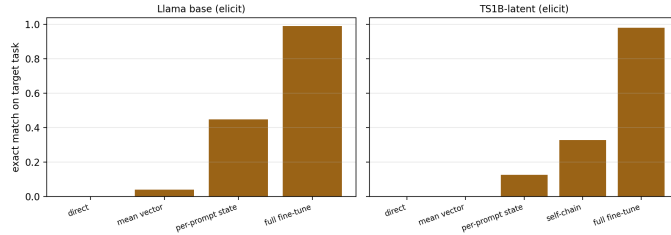


Figure 4: Intervention ladder on the elicit parent (weights frozen except the last bar).

**Metric.** Relative write  $\|\Delta W\|/\|W\|$  (exact from LoRA factors); total travel and per-step speed from adapter snapshots; capability per unit travel = final EM / total travel.

**Result.** Table 5, Figure 5. Relative write 0.212 (teach) vs 0.095 (elicit),  $2.2\times$ ; total travel 457 vs 120,  $3.8\times$ , over  $2\times$  as many steps (23.5K vs 11.5K); capability per unit travel  $2.0 \times 10^{-4}$  vs  $8.1 \times 10^{-3}$ ,  $\sim 40\times$ .

**Elicit vs. teach.** Eliciting connects what exists; teaching writes what does not. The temporal *shape* of writing (burst then decay) is shared by both arms — an optimisation signature — so the discrimination is in amplitude, duration, and efficiency, not timing.

## R8 The regime difference is total: the two children differ at both node and edge level.

**Metric.**  $\Delta S = 1 - \text{Jaccard}$  at node level (@32) and edge level (@256, node→block edge attribution).

**Result.** Elicited @1M vs taught @1M:  $\Delta S_{\text{node}} = 0.769$ ,  $\Delta S_{\text{edge}} = 0.739$  (ratio 0.96), both far above the noise floors (0.32 / 0.61).

**Elicit vs. teach.** When the mechanism differs, it differs everywhere — membership and wiring alike — consistent with R3: not a re-routed version of one machine but two machines.

| Patient + intervention (k=32 circuit nodes) | condition             | format | EM           |
|---------------------------------------------|-----------------------|--------|--------------|
| TS1B-latent + child mean shift              | circuit nodes         | 0.164  | 0.000        |
| TS1B-latent + child per-prompt states       | circuit nodes         | 0.359  | <b>0.125</b> |
|                                             | random-32 nodes       | 0.051  | 0.000        |
| TS1B-latent self-chain (no donor)           | own rewrite + “= ”    | –      | <b>0.328</b> |
| blank + binding dose, self-chain            | parser without engine | –      | 0.000        |
| blank + taught donor                        | every condition       | 0.000  | 0.000        |

Table 4: Zero-training interventions, TinyStories twins.

|                                          | elicit (TS1B-latent) | teach (blank)        |
|------------------------------------------|----------------------|----------------------|
| relative write $\ \Delta W\ /\ W\ $ (1M) | 0.095                | 0.212                |
| steps to convergence                     | 11,500               | 23,496               |
| peak / final writing speed (per step)    | 0.218 / 0.018        | 0.306 / 0.022        |
| total weight travel $\ \Delta W\ _F$     | 120                  | 457                  |
| final exact match                        | 0.981                | 0.093                |
| capability per unit travel               | $8.1 \times 10^{-3}$ | $2.0 \times 10^{-4}$ |

Table 5: Weight-space write, elicit vs teach.

## Validation results (not regime differences, but what licenses R2–R6)

- **Both circuits are real and load-bearing.** True activation patching: elicited child top-8 of 528 nodes = 0.997 sufficient / 0.998 necessary (top-32: 0.999 / 1.000); taught child top-8 necessity 0.986, top-32  $\sim$ 0.999. Every overlap above compares provably load-bearing sets.
- **Construction controls.** Identical doses on the blank twin change nothing on the target (0.000 in every probe); a sum-only binding dose leaves “difference” unbound (per-word binding); the composition law chain = parse  $\times$  compute held on four variants.

## Informative non-discriminators (measured, reported for completeness)

- Effective rank of  $\Delta W$  (participation ratio): elicit 7.1, teach 5.2 of 512 — both energy-concentrated; rank does not separate the regimes. Alignment of  $\Delta W$  with the base’s top-64 singular directions is near the 0.058 baseline for both.
- Temporal profile of weight travel: burst-then-decay in both arms (R7).
- Prompt brittleness (16-shot collapse) is a property of this lineage’s training-sequence structure, not of the regime: both children score  $\sim$ 0 at 16 shots.
- Edge-level  $\Delta S$  within the elicit arm: the edge instrument’s split-half noise floor (0.61–0.62) exceeds the measured churn on both eval surfaces — a sensitivity limit.

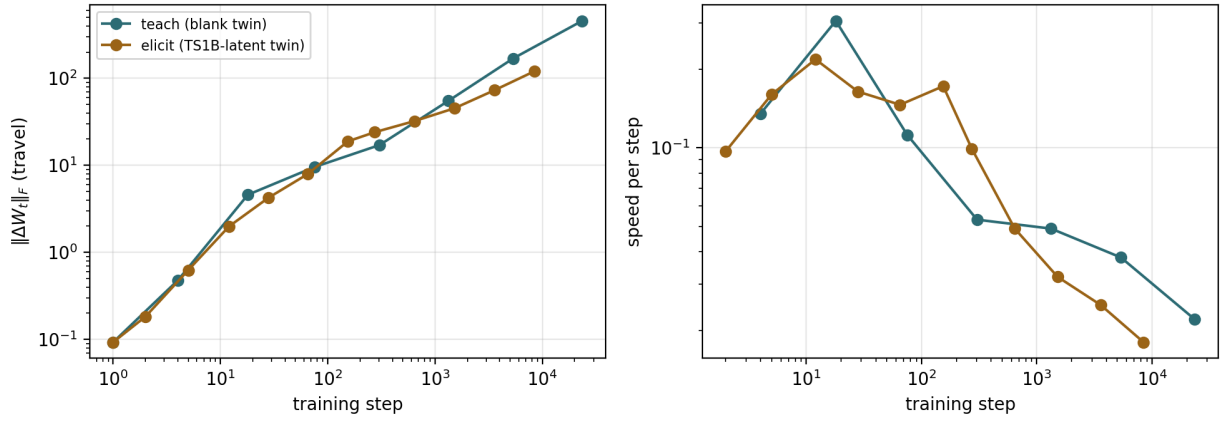


Figure 5: Weight travel (left) and speed (right) from adapter snapshots; profiles coincide, amplitudes do not.